

Lesson 9 — When the Sensor Talks to the Motor

Matches LEGO Teacher Portal lesson: "Ride Your Way".

Objective (I can...): I can make the color sensor and the motor work together — the color decides what the motor does. **CS concept:** How hardware parts interact — a color reading controls the motor · **Builds on:** Lesson 8 (color sensor) and Lesson 6 (motor).

AI Connection (2 min, read aloud): In a real robot, the "eyes" tell the "wheels" what to do — sensor talks to motor. AI robots do this all day. *Ask the group:* "If the sensor sees red, what would be a good thing for the motor to do?"

Materials: LEGO CS & AI kit + device per pair, color cards (red = stop, green = go). **Model reuse:** Hardware machine from Lesson 6 — code only.

60-minute flow - Warm-up (10): "Red light, green light" with bodies — green card, everyone walks; red card, freeze. The color is telling the "motors" (legs) what to do. - **Concept + demo (8):** Demo: when the sensor sees green, the motor spins; when it sees red, the motor stops. Hold up each card and show the motor obey. - **Build/Code (25):** Pairs code the sensor-to-motor link: green seen → motor go; red seen → motor stop. They test with real cards. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a third color that makes the motor go slow or backward. - **Share + tidy-up (10):** Pairs run their "traffic motor" with the cards and say which color does what. Model stays built.

Pair roles: Driver codes, Navigator works the color cards and watches the motor. Swap at the wiggle break. (*Trio rotates code/cards/watch.*)

Differentiation: *Support* — just two colors, go and stop. *Extension / fast-finisher* — three colors doing three different motor moves. *Reteach* — do green-go alone first, confirm it works, then add red-stop.

Exit check: Pair's motor changes what it does based on the color the sensor sees. **At-home (optional):** Watch a real traffic light; tell a grown-up what cars do at each color. ~5 min.
